



Airborne small-footprint discrete-return LiDAR data in the assessment of boreal mire surface patterns, vegetation, and habitats

Ilkka Korpela^{*}, Markku Koskinen, Harri Vasander, Markus Holopainen, Kari Minkkinen

Faculty of Agriculture and Forestry, University of Helsinki, P.O. Box 27, University of Helsinki, FI-00014 Helsinki, Finland

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ABSTRACT

Boreal mires encompass high diversity in species and habitats, many of which are endangered. In Finland, extensive use of peatlands has resulted in habitat fragmentation. A need for accurate and cost-efficient vegetation mapping and monitoring of habitat changes exists in mire conservation, restoration and peatland forestry. LiDAR is an emerging and excellent tool for probing the geometry of vegetation and terrain. Modern systems measure the backscattered signal accurately and also provide radiometric information. Experiments were carried out in a complex minerotrophic–ombrotrophic eccentric raised bog in southern Finland (61°47'N, 24.18'E). First, we tested discrete-return LiDAR for the modeling of mire surface patterns and the detection of hummocks and hollows, as well as the effect of mire plants on the Z accuracy of the surface echoes. Secondly, the response of different mire vegetation samples in LiDAR intensity was examined. Thirdly, we tested area-based geometric and radiometric features in supervised classification of mire habitats to discover the meaningful variables. The vertical accuracy of LiDAR in mire surface modeling was high: 0.05–0.10 m. A binary hummock-hollow model that was estimated from a LiDAR-based elevation model matched flawlessly in aerial images and had moderate explanatory power in habitat classification trials. The intensity of LiDAR in open-mire vegetation was mainly influenced by the surface moisture, and separation of vegetation classes spanning from ombrotrophic to mesotrophic vegetation proved to be difficult. Area-based features that characterize the height distribution of LiDAR points in the canopy were the strongest explanatory variables in the classification of 21 diverse mire site types. Actual qualifying differences in the ground flora were unattainable in the LiDAR data, which resulted in inferior accuracy in the characterization of ecohydrological conditions and nutrient level of open mires and sparsely forested wet sites. Mire habitat classification accuracy with LiDAR surpassed earlier results with optical data. The results suggested that LiDAR constitutes an efficient aid for monitoring applications. We propose the co-use of images and LiDAR for enhanced mapping of open mires and tree species. *In situ* calibration and validation procedures are required until invariant geometric and radiometric features are discovered.

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1. Introduction

Airborne LiDAR is an active remote sensing technique that measures the properties of scattered light to determine the range and other information of a distant target (Wehr and Lohr, 1999). LiDAR is an efficient tool for probing vegetation geometry and topographic features. In the last 4–8 yrs, it has revolutionized photogrammetry and environmental 3D data acquisition. LiDAR is particularly suitable in forests, because it enables coincident measurement of the vegetation and terrain, which cannot be done reliably in passive images. LiDAR is less dependent on weather conditions than optical sensing. Prevalent method is the use of laser pulses, with frequencies in excess of 150 kHz. LiDAR provides

both geometric and radiometric data. Discrete-return LiDAR sensors extract 1–4 returns per pulse, 3D points, and intensity values that correspond to backscatter amplitude. Full-waveform LiDAR samples at GHz rate the entire reflected waveform for computer-intensive postprocessing and extraction of points and elaborate waveform features (Wagner et al., 2006). LiDAR provides discrete point sampling, whereas images have a continuous 2D coverage. LiDAR is less affected by shadows and occlusions, which constitutes an advantage over optical images. The XY accuracy of the pulse center is typically 0.1–0.5 m, depending on the flying height. The accuracy in Z is usually better than 0.2 m. The footprint, i.e. the illuminated area on the ground, is determined by the beam divergence. Values range from 0.2 m to 1.0 m for flying heights of 1–5 km.

Airborne LiDAR is a powerful technique to use in terrain modeling even under leaf-on conditions (Kraus and Pfeifer, 1998; Korpela and Välimäki, 2007). The canopy returns can be used to

^{*} Corresponding author. Tel.: +358 400 218305; fax: +358 9 19158159.

E-mail address: ilkka.korpela@helsinki.fi (I. Korpela).

characterize forest vegetation height, type and density (Næsset, 2004). Co-use of images improves species estimation accuracy, which in turn enhances the accuracy of the biomass estimates (Packalén and Maltamo, 2007; Korpela et al., 2008). For monitoring applications, the repeatable and high absolute XYZ accuracy is advantageous, since changes can be detected at submeter scales (Yu et al., 2006, 2008) and the same measurement units, e.g. trees, plots or 3D voxels, can be monitored over time. This reduces the effect of sampling error in the change estimates.

Using LiDAR to obtain 3D data of objects is simple and cost-effective and explains why LiDAR is replacing photogrammetry in many tasks, although 3D reconstruction and object classification still remain a challenge regardless of the method by which the 3D point data were obtained. In discrete-return LiDAR, the intensity carries radiometric information to support object classification. A disadvantage for users is that the intensity is not affected only by the target backscatter reflectance. Factors that influence intensity also include the intentional and random power variation of the transmitted pulse, the speckle effect of the laser, electronic noise in the receiver and the background noise, changes in the receiver sensitivity, the wavelength used, surface geometry, pulse width, scanning distance and angle, and two-way transmission losses in the path of the signal (Wehr and Lohr, 1999; Ahokas et al., 2006; Kaasalainen et al., 2007a; Morsdorf et al., 2007). The size of the footprint influences the XYZ and intensity distributions in vegetation canopies (Hopkinson, 2007). However, it should be stressed here that LiDAR intensity cannot be used for deriving the reflectance of shoots, leaves and needles that are normally smaller than the illuminated area (Wagner et al., 2008). In practice, the intensity variation caused by the varying LiDAR-to-target range is the most important factor to be accounted for in a LiDAR campaign (Hopkinson, 2007; Kaasalainen et al., 2007a; Höfle and Pfeifer, 2007; Korpela, 2008). This reduces noise in the intensity data. However, the normalization for range is ambiguous for targets such as the forest canopy, because the normalization coefficient depends on the size of the scatterers (Ahokas et al., 2006; Korpela, 2008; Wagner et al., 2008). Work on developing methods for absolute calibration and retrieval of surface albedo from calibrated LiDAR observations is also ongoing (Kaasalainen et al., 2005, 2007a; Ahokas et al., 2006; Wagner et al., 2008). Even in calibrated data, retrieval of the reflectance of objects smaller than the footprint remains ill-posed. This constitutes a challenge for the use of LiDAR radiometry in vegetation classification and explains why the co-use of images is sought in many cases (Koetz et al., 2008; Korpela et al., 2008). Currently, topographic sensors are used in forest applications. These operate on a single wavelength, typically in the near-infrared band, and are optimized for terrain modeling. Future sensors are anticipated to be multispectral, which may increase the range of applications (Kaasalainen et al., 2007b).

Intensity has not been used extensively for vegetation mapping. One reason may be that intensity measurements have been noisy until recently, since sensor manufacturers first focused on improving range determination accuracy. Intensity metrics were used for the classification of tree species (Holmgren and Persson, 2004; Donoghue et al., 2007; Brandtberg, 2007; Ørka et al., 2007), for modeling of canopy gap fraction (Hopkinson and Chasmer, 2007), and for land-cover classification (Brennan and Webster, 2006; Yoon et al., 2008). Korpela (2008) examined range-normalized intensity data for the mapping of ground lichens. Normalization improved the precision of the classifications.

Mire habitats occupy significant proportions of the Boreal Zone, in Fennoscandia, Canada, and Russia. In Finland, drained and pristine mires constitute 28% of the land cover (Korhonen et al., 2008). In southern Finland, 75% of mires have been drained for forestry and agriculture (Virkkala et al., 2000). Mires are also utilized for peat harvesting for energy and horticultural purposes.

In Finland, restoration of drained mires was begun in the 1990s (Aapala et al., 2008). Boreal mire ecosystems are sensitive to climate changes and changes in the vegetation and canopy structure due to atmospheric nutrient fallout (Strack, 2008). Efficient monitoring tools for change detection and the appraisal of restoration success are needed. Currently, mire habitats are mainly mapped using field surveying, in which aerial images are a stratification aid, but the work is laborious and expensive.

The use of LiDAR or other 3D remote-sensing methods in mire vegetation appraisal is largely unexplored. Waser et al. (2008) used a method that combines advanced image matching of large-scale aerial images with sparse leaf-off LiDAR for the detection of changes in forested area and shrub encroachment in prealpine mire environment. Reliable 3D reconstruction of canopies by stereo matching proved imprecise, due to occlusions. Large image overlaps are required, which leads to multi-image matching (e.g. Hirschmugl et al., 2007). Underestimation of tree heights and imprecision of canopy surface models acquired by means of stereo image matching were also reported (Næsset, 2002; Korpela and Anttila, 2004). The use of interest operators for finding corresponding treetops in multiple images for 3D treetop positioning leads to accurate height determination (Korpela, 2007), but this approach has not been tested in mire environments, where crowns are often seen in low contrast and trees exhibit considerable height variation. In view of the difficulties of image matching, LiDAR is a very attractive option for the reconstruction of canopies under the conditions found in mires. Vehmas et al. (2008, 2009) used airborne LiDAR in site-type mapping of mineral soil forests in eastern Finland, especially for the detection of lush herb-rich sites. The detection was based on the characteristics of point-height distributions in the canopy. The site types correlated with the height growth rate of trees; however, the site types were defined by the understory vegetation: the occurrence of certain herbs, ferns, and bushes. Gatzliolis (2007) estimated site indices in the coastal Pacific Northwest of the USA, using LiDAR measurements of dominant height.

Images and their monoscopic interpretation are often the only options in retrospective vegetation analysis, since commercial LiDAR has been available only since 2002–2004. Historical and up-to-date aerial photographs and hyperspectral data were used in peatland habitat mapping in Finland by Holopainen (1998). The habitat classes that were sought were defined according to drainage conditions, main tree species, degree of stocking, and site fertility—features that would be measurable in optical images but still relevant for ecological decision-making. Holopainen (1998) generalized 21 pristine mire site types into nine habitats. The most fertile mineral soil classes were difficult to detect, while treeless mire habitats were successfully separated in optical data.

Our hypotheses are that airborne LiDAR data, through their proven ability to probe the geometric and radiometric properties of all vegetation layers, could be used for retrieving the key taxonomic features of pristine mire site types in Finland. It would be suited for the long-term monitoring of changes in mire surface structure and vegetation. More precisely, we see the potential for (a) accurate digital elevation modeling (DEM) and retrieval of the hummock-hollow mire surface patterns and (b) for measurement and characterization of the density, vertical distribution, and species mixture of ground, shrub layer, and tree vegetation.

- We carried out experiments on two nearby mires and tested,
- the geometric accuracy of LiDAR in retrieving the mire surface elevation,
- the response of LiDAR intensity to different mire vegetation surfaces, and
- LiDAR- and DEM-based features in the classification of pristine mire site types, mire habitats, dominant tree species, and the nutrient status of the site types.



Fig. 1. 1.1 km × 2.2 km orthoimages of Lakkasuo mire from 1946 (left, pristine), 1963, and 2002 (right). Rectification was done using the LiDAR DEM of 2006 (Fig. 2).

2. Test sites and LiDAR data

2.1. Lakkasuo mire complex

Lakkasuo mire is an eccentric raised bog in southern Finland (61°47'N, 24.18°E, 145–160 m a.s.l.) (Fig. 1). It is a diverse mire complex containing a large proportion of mire site types in southern Finland extending over an area of 1 km × 2 km. In 1961, 40% of Lakkasuo was drained for forestry. Ombrotrophic sites constitute 56% of the pristine part and the remainder is minerotrophic, fed by springs and subsurface runoff (Fig. 2). The pristine mire types include all three main types: treeless mires,

genuine forested mire types and sparsely forested composite site types (Laine and Vasander, 1996). In all, 21 mire site types are found in Lakkasuo (Table 1). The classification is botanical and the mire site type, i.e. the structure and species composition of the vegetation community, represents the primary qualities of the habitat (*sensu* Cajander, 1913). The simultaneous variation in ecohydrology and nutrient level influences the vegetation communities in pristine mires, resulting in a manyfold higher number of mire site types in comparison to site types in mineral soils. Drained peatlands lack variation in ecohydrologic conditions and their postdrainage succession leads to seven site types that are somewhat similar to those on mineral soils (Vasander and Laine, 2008). There are different ecological divisions of mire site types. We used partitioning according to the main tree species, nutrient level, and the three main types (Table 1). Genuine forested mires are characterized by hummock- and intermediate-level vegetation and they are usually drier than treeless mires and composite types, which consist of flark- or lawn-level vegetation. Pine (*Pinus sylvestris* L.) is the main tree species in Lakkasuo and is the only species on the most barren ombrotrophic sites, which receive nutrients only from rainfall. Spruce (*Picea abies* (L.) H. Karst.) and downy birch (*Betula pubescens* Ehrh.) are minerotrophic species. Alder (*Alnus glutinosa* (L.) Gaertner) is found at the richest sites. A thorough description of Lakkasuo is found in Laine et al. (2004).

In 1995, Lakkasuo was mapped using a systematic 50 m × 50 m grid of plots (Larmola, 1997; Vähäsarja, 1997). The site type was determined in addition to measurements on the peat chemistry, water table and tree biomass. We used the 11-yr-old data and a visual interpretation of aerial photographs from 2002 to draw a reference site-type map (Fig. 2, Table 1). A block of 60 oriented aerial images from 1946 to 2007 was available for Lakkasuo (Korpela, 2006).

The LiDAR campaign of Lakkasuo occurred on July 25, 2006. Precipitation 4 weeks prior to the campaign was only 37 mm in the area. The dry season may have affected our results, since reflectance at $\lambda = 1064$ nm is sensitive to surface moisture. An ALTM3100 sensor was flown 1.2 km above the mire and operated at a pulse repetition frequency of 70 kHz. The mirror frequency was 42 Hz and the scan angle $\pm 14^\circ$. The sensor registered 1–4 returns with intensity in eight effective bits. The pulse density was 1–3 m⁻² (Fig. 3) and footprint

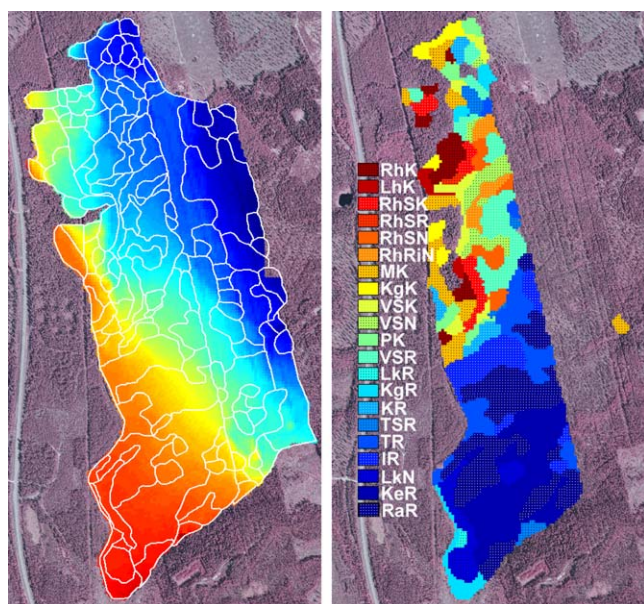


Fig. 2. Left: LiDAR DEM (145–160 m) of Lakkasuo also covering the drained areas. The white lines delineate 144 visually interpreted site-type compartments in the coverage of the field plot grid. Right: Pristine mire site types, 67.05 ha. The ombrotrophic bogs (TR, IR, LkN, KeR, RaR) predominate in the south. Minerotrophic mire site types are found in the north.

Table 1

Code, name, main type, nutrient level, tree species mixture and area of the 21 pristine mire site types in Lakkasuo. The site types are listed in decreasing order of nutrient level. Descriptions of the site types are found in Laine and Vasander (1996, 2008).

Code		Nutrient level	Tree species	Area (ha)
Genuine forested types				
RhK	Herb-rich hardwood–spruce mire	meso-eu	S–B	2.8
LhK	Eutrophic paludified hardwood–spruce mire	meso-eu	S–B	0.7
MK	Vaccinium myrtillus spruce mire	meso	S	4.0
KgK	Paludified spruce mire	meso	S	1.7
PK	Vaccinium vitis-idaea spruce mire	meso-oligo	S–P	1.5
KgR	Paludified pine mire	meso-oligo	P	2.0
KR	Spruce–pine mire	meso-oligo	S–P	1.6
IR	Dwarf shrub pine bog	oligo-ombro	P	4.5
RaR	Sphagnum fuscum pine bog	ombro	P	12.8
Sparsely forested composite site types				
RhSK	Herb-rich sedge hardwood–spruce fen	meso	S–A–B	2.1
RhSR	Herb-rich sedge birch–pine fen	meso	P–B	0.2
VSK	Tall-sedge hardwood–spruce fen	oligo-meso	S–B	2.1
VSR	Tall-sedge pine fen	oligo-meso	P–B	5.3
LkR	Low-sedge Sphagnum papillosum pine fen	oligo-ombro	P	0.2
TSR	Cottongrass-sedge pine fen	oligo-ombro	P	1.2
TR	Cottongrass pine bog	ombro	P	5.4
KeR	Ridge-hollow pine bog	ombro	P	10.0
Treeless mires				
RhSN	Herb-rich sedge fen	meso	–	1.8
RhRiN	Herb-rich flark fen	meso	–	1.6
VSN	Tall-sedge fen	oligo-meso	–	2.5
LkN	Low-sedge bog	ombro	–	3.1

Nutrient level: eu-, meso-, oligo-, ombrotrophic; classes of forested types indicate also the yield, which is $0.5\text{--}4\text{ m}^3\text{ ha}^{-1}\text{ a}^{-1}$ in Lakkasuo. Tree species: P = Scots pine, S = Norway spruce, B = downy birch, A = alder.

0.36 m. Blom Geomatics AS (Oslo, Norway) did the preprocessing in which five strips were matched to remove errors in the trajectory and attitude data and TerraModeler (Terrasolid Ltd., Jyväskylä, Finland) was used for the classification of ground points. We then created a raster DEM at 1-m resolution (Fig. 2) from the ground points, using triangulation algorithms by Isenburg et al. (2006).

The proportion of pulses with 1, 2, 3, or 4 returns was 73.2%, 24.1%, 2.6%, and 0.1%, respectively. Because of the minimum range

discrimination distance of 3 m of the ALTM3100 and the prevailing low vegetation, pulses with 1 or 2 returns predominate (Fig. 3). The geometric accuracy of the LiDAR was assessed to be 0.3 m or better in XY and better than 0.1 m in Z. The XY assessment was done by superimposing the LiDAR points on large-scale aerial images with narrow linear features such as road markings and power line cables (cf. Korpela, 2006). We derived the Z estimate from the error estimate of the LiDAR trajectory and from four Network RTK

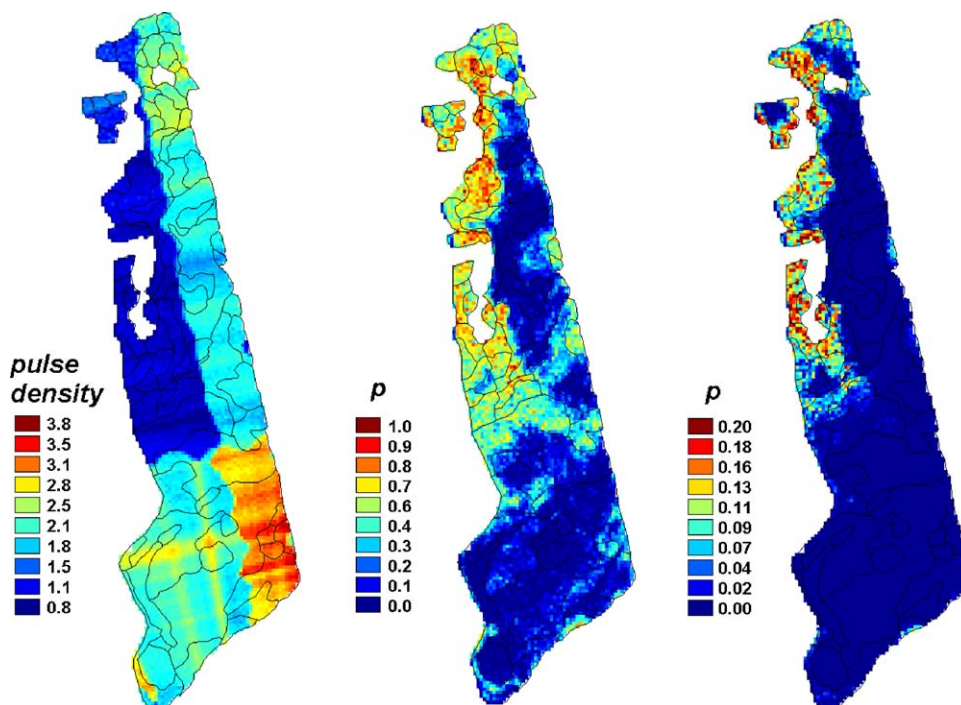


Fig. 3. Left: Pulse density $0.9\text{--}3.8\text{ p/m}^2$. The large-scale density variations are due to strip overlaps (1, 2, or 3) and the sudden undulations were caused by variation in the airplane's pitch-angle. Middle: Proportion of pulses with two or more returns. Right: Proportion of pulses with three or four returns. The maps are drawn in a 10-m grid and cover the pristine part of Lakkasuo mire, 67.05 ha.

Table 2Description of the 0.25-m² vegetation samples in the Silmäpäänlammi fen.

Class		n	Species composition, '+' = abundant
1	RaN	45	<i>Sphagnum fuscum</i> (Schimp.) Klinggr., <i>Sphagnum rubellum</i> Wils., <i>Sphagnum magellanicum</i> Brid., <i>Sphagnum angustifolium</i> (Russ.) C. Jens., <i>Eriophorum vaginatum</i> L., <i>Andromeda polifolia</i> L., <i>Carex loliacea</i> L., <i>Rubus chamaemorus</i> L.
2	LkN	27	<i>S. angustifolium</i> , <i>S. magellanicum</i> , <i>E. vaginatum</i> (+), <i>A. polifolia</i>
3	LkN-VSN	58	<i>S. magellanicum</i> , <i>Sphagnum fallax</i> (Klinggr.) Klinggr., <i>E. vaginatum</i> , <i>Carex rostrata</i> Stokes, <i>Carex magellanica</i> Lam.
4	VSN	54	<i>S. fallax</i> , <i>C. rostrata</i> , <i>Menyanthes trifoliata</i> L.
5	RhSN	13	<i>S. fallax</i> , <i>C. rostrata</i> (+), <i>Comarum palustre</i> L., <i>Eriophorum angustifolium</i> Honck.
6	SPHFAL	8	<i>S. fallax</i>
7	FUSHUMA	15	<i>S. fuscum</i> hummock-A; <i>Polytrichum strictum</i> Brid., <i>S. rubellum</i> , <i>Empetrum nigrum</i> L., <i>Rhododendron tomentosum</i> Harmaja, <i>E. vaginatum</i> , <i>Vaccinium uliginosum</i> L.
8	FUSHUMB	18	<i>S. fuscum</i> hummock-B; <i>P. strictum</i> , <i>E. nigrum</i> , <i>E. vaginatum</i>
9	PEAT	13	Peat surface without vegetation (mud-bottom)
10	RALBA	13	<i>Rhynchospora alba</i> (L.) Vahl
11	CALPUR	1	<i>Calamagrostis purpurea</i> (Trin.) Trin.
12	MAGEL	5	<i>S. fallax</i> , <i>C. magellanica</i>
13	SPHFUS	1	<i>S. fuscum</i>
14	ANDR	15	<i>S. fuscum</i> , <i>S. rubellum</i> , <i>A. polifolia</i>
15	MENYTR	8	<i>S. fallax</i> , <i>M. trifoliata</i> ,
16	CLIMOS	8	<i>S. fallax</i> , <i>Carex limosa</i> L.
17	DROS	2	<i>Drosera anglica</i> Huds. (mud-bottom)

(Wanninger, 2005) observations made in Lakkasuo. The accuracy of the DEM was, however, not verified in the field. It was undoubtedly influenced by the variations in pulse density and vegetation density. In seedling stands 3–6 km from Lakkasuo, a DEM derived from data acquired on the same day using the same sensor (850 m, 6 pulses per m²) showed a standard deviation of 0.19 m, when DEM values were compared with Network RTK (Korpela et al., 2008). Soil preparation had taken place on these sites and there was 1–6-m-high dense vegetation. We hence assumed that the Lakkasuo-DEM had a RMS-accuracy of 0.05–0.15 m being highest in the open and sparsely forested areas and in areas of high pulse density (Fig. 3).

2.2. Silmäpäänlammi fen

Samples of open-mire vegetation surrounding a pond under the process of terrestrialization were collected on July 12–13, 2006. For positioning we used a Network RTK system that provides an accuracy of 2–5 cm in XYZ. The samples consisted of vegetation typical of five mire site types, classes 1–5 in Table 2. In addition, samples of characteristic mire vegetation and hummocks in the treeless part of the mire were positioned (Table 2, Fig. 4). The samples were 0.5 m × 0.5 m in size. The LiDAR data of Silmäpäänlammi, 8 km north of Lakkasuo, were acquired 1 h before



Fig. 4. Vegetation samples around the Silmäpäänlammi pond superimposed in a color-infrared aerial image that was taken August 2, 2006 at 0630 GMT. Vexcel UltraCAM D camera and 1-km flying height. The samples cover an area of 130 m × 100 m. The direction of the flight line was towards the south.

Lakkasuo from a flying height of 850 m. It had three overlapping strips and the pulse density was 6–9 pulses per m². The footprint was 0.26 m.

3. Experiments

For the photogrammetric and LiDAR analyses we used a photogrammetric workstation written in Visual Basic and C languages, as well as routines written in Python language. Statistical analysis and classifications were done using R (version 2.8.1).

3.1. LiDAR in mire elevation and surface structure modeling

We extracted the LiDAR data in 0.5-m-wide circles centered at the sample point locations at the Silmäpäänlammi fen. The closest points ($n = 304$) were 5.8 cm above the reference heights ($SD = 6.4$ cm). We examined the effect of vegetation by comparing the vegetation classes. In classes 1, 6, 9, and 13, which represent pure moss and peat surfaces, the overestimation of Z was 2.7–5.4 cm. In classes 4 and 5, which had 20–50-cm-high sedge vegetation, the differences were 6.0 cm and 6.6 cm, respectively. Classes 7 and 8 (hummocks with shrub vegetation) showed differences of 5.7 cm and 8.0 cm, respectively. The points in samples with *Menyanthes trifoliata* were 5 cm higher in comparison to pure *Sphagnum fallax* moss growth, class 16 vs. 6. In all, the mire vegetation caused a bias of 2–5 cm, which is almost negligible in DEM estimation. The observed difference of 3–5 cm for pure moss surfaces may partly be explained by systematic errors in the field measurements as the peat surface subsided near the GPS surveyor. The LiDAR data may also have contained systematic Z errors. Changes in the water level between July 12 and 25, 2006 cannot be excluded. If the precision of the reference data were 5 cm and if it were excluded from the observed variance, the precision of LiDAR in measuring the surface elevation was approximately 5 cm. This suggests that *Sphagnum* surfaces

constitute well-defined reflectors for LiDAR and that in high-density, small-footprint data with high XY and Z accuracy the mapping of the surface details seems achievable in open and sparsely forested sites, as illustrated in Fig. 5.

The hummock-hollow structure of the mire surface is characteristic of many of the pristine mire site types. Encouraged by the results at Silmäpäänlammi, we attempted to derive the hummock-hollow structure of the Lakkasuo mire, using the LiDAR DEM. The hummocks and hollows were defined as local elevation deviations in the surface. Because of the elevation gradients in Lakkasuo, the hummocks were defined inside 20 m × 20 m areas and a plane was fitted to the DEM data to remove possible local trends in elevation. The sign of the residuals was used in labeling a binary version of the DEM (the hummock model) (Fig. 6). The SD and range of the residuals were also stored. The procedure was done in a 10-m grid.

The hummock model was used for deriving features for the determination of mire site types, using supervised classification (Section 3.3). They were computed inside 400-m² windows and consisted of the SD of the elevation residuals (*sdz*), range of elevation residuals (*rangez*), number and mean size of hummocks (*humnum*, *hummsize*), and a compactness measure of the hummocks (*humcomp*) that was computed from the number of border pixels (a hummock pixel with hollow pixels in the 8-neighborhood). A simple algorithm for 8-connected components labeling was used in determining the unique hummocks in each window.

3.2. Analysis of LiDAR intensity in different open-mire vegetation types

The raw intensity DN data were normalized for the varying range (*R*) to reduce noise:

$$INT_{norm} = INT_{raw} \cdot \left(\frac{R}{R_{ref}} \right)^{2.5}, \quad (1)$$

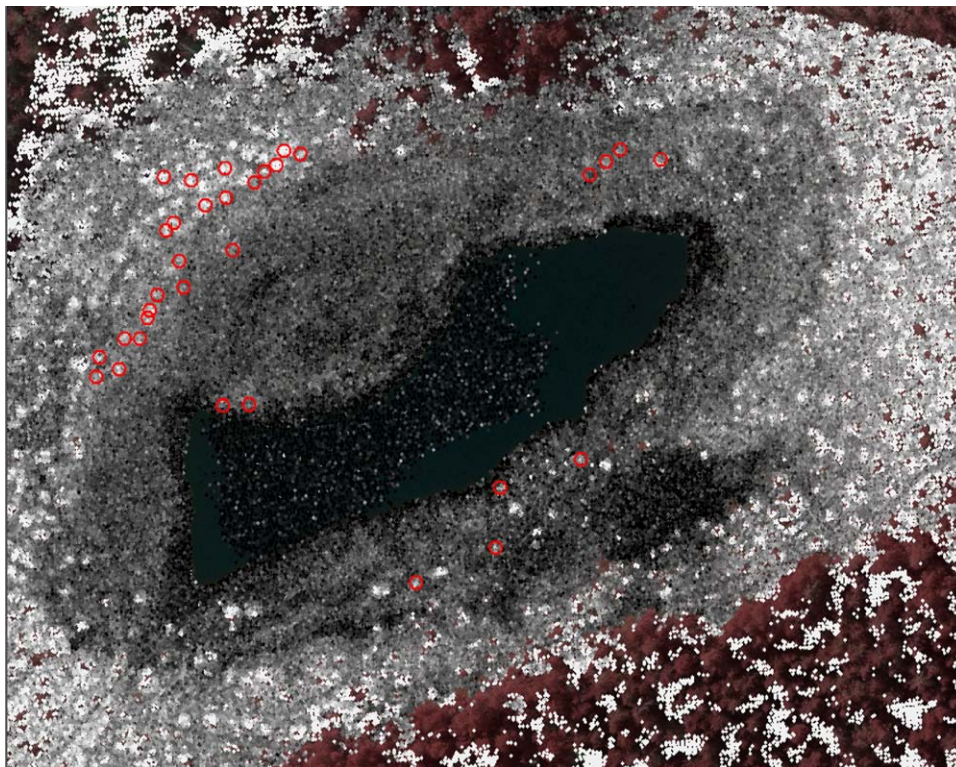


Fig. 5. Near-ground LiDAR points superimposed in an aerial image (Fig. 4). The grayscale denotes elevation, 181.8–182.3 m. Positions of hummocks ($n = 33$) are marked with the red circles. The patterns match well and LiDAR seems able to detect the 15–30-cm-high, 1–2-m-wide hummocks in the open mire. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

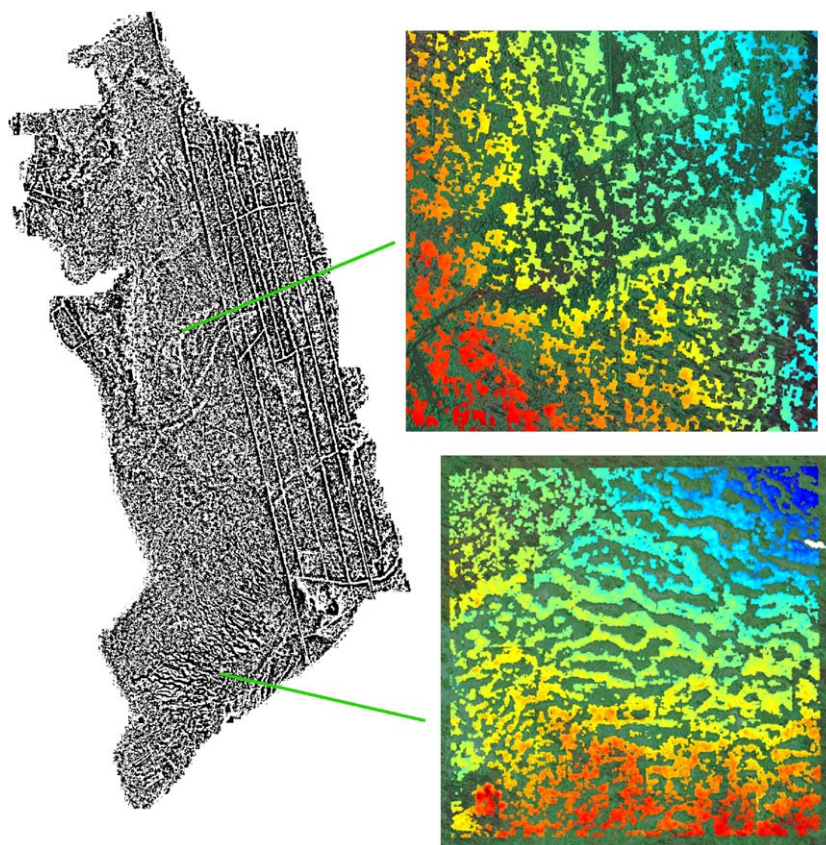


Fig. 6. Hummock model of Lakkasuo mire. The 1961 network of ditches is visible in the east. The patterned ridge-hollow pine bog is seen in the south. It has elongated hummocks aligned in parallel with the elevation contours. The hummock model is superimposed in aerial images in the herb-rich flark fen (upper) and in the ridge-hollow pine bog (lower). The colors depict elevations given by the DEM.

where R_{ref} was 1240 m at Lakkasuo and 809 m at Silmäpäänlammi. The low-altitude campaign covered parts of Lakkasuo, but we made no attempts to use it, since the LiDAR settings were different. The power factor of 2.5 in (1) was initially determined for understory vegetation in nearby pine forests (Korpela, 2008). It is a compromise. The value is theoretically 2 for large surfaces, 3 for linear scatterers, and 4 for objects smaller than the footprint. Thus a different value would have been needed for canopy and mire surface points. The distribution statistics of INT_{norm} for the vegetation samples in Silmäpäänlammi fen are in Table 3. The SDs were 10–15% larger for raw intensity data. Range normalization thus reduced intraclass variation.

Table 3

Mean and SD of INT_{norm} in classes with more than seven field samples. Silmäpäänlammi fen.

Class		Mean	SD
1	RaN— <i>S. fuscum</i> bog vegetation	94.5	7.0
2	LkN—Cottongrass bog vegetation	97.1	9.2
3	LkN—VSN—Low-sedge fen vegetation	90.8	12.1
4	VSN—Tall-sedge fen vegetation	94.7	13.2
5	RhSN—Herb-rich sedge fen vegetation	102.0	11.6
6	SPHFAL— <i>Sphagnum fallax</i>	90.0	8.1
7	FUSHUMA—Hummock	98.8	7.5
8	FUSHUMB—Hummock with <i>Rhododendron tomentosum</i>	96.0	7.7
9	PEAT—mud-bottom	51.2	5.8
10	RALBA— <i>Rhynchospora alba</i>	64.3	8.4
14	ANDR— <i>Andromeda polifolia</i> with <i>S. fuscum</i>	75.8	13.0
15	MENYTR— <i>Menyanthes trifoliata</i> with <i>S. fallax</i>	89.3	9.7
16	CLIMOS— <i>Carex limosa</i> with <i>S. fallax</i>	73.5	10.8

Fig. 7 shows the intensity map of the ground returns in Silmäpäänlammi.

It seems plausible that target wetness exerted a major effect on intensity (Fig. 7, Table 3). Low values were observed near the shores of the pond. Class 9 represents vegetation-free and wet peat and it had the lowest intensity. These surfaces were found along the shores of the pond, where very dense, 10–20-cm-high tussocks of *Rhynchospora alba* were also present and showed low intensity despite their vigor. Vegetation of class 14, *Andromeda polifolia*–*S. fuscum*–*S. rubellum*, is very similar to the class 1 *S. fuscum* bog with the exception that samples in class 14 were all located near the pond, which explains the differences in mean intensity (75.8 vs. 94.6).

The highest intensity was observed in class 5, herb-rich sedge fens. It can be explained by the presence of the herbaceous species *M. trifoliata* and *Comarum palustre*. However, a large part of the observed interclass intensity variation may have been due to differences in mire surface moisture, e.g. as indicated by the difference in classes 6 and 16, where the same peat moss, *S. fallax*, resulted in lower intensity when *Carex limosa*, an indicator of particularly wet conditions, was present. Classes 6 and 16, pure *S. fallax* and *S. fallax* with *Menyanthes trifoliata*, did not deviate from each other, although *Menyanthes* could be expected to give rise to high intensity because of the favorable orientation of the relatively large leaves. The mire site-type vegetation of classes 1–5 could not be reliably separated, based on the intensity data (Table 3), which implies that the usability of intensity can be restricted in the characterization of oligo-mesotrophic vegetation.

The range-normalized intensity ($\lambda = 1064$ nm) correlated with the NIR ($\lambda = 670$ – 940 nm) channel of an aerial image except for shaded targets, where the NIR values in images were low (Fig. 4).

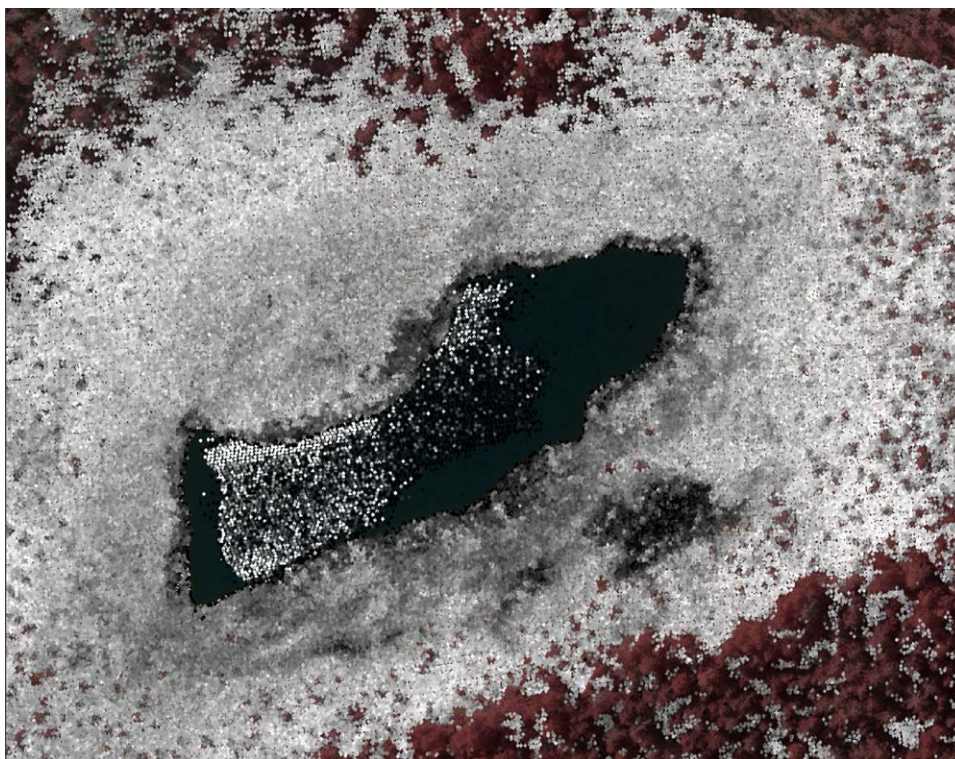


Fig. 7. LiDAR intensity map of ground points (<0.3 m) in Silmäpäänlammi (cf. Figs. 4 and 5). Grayscale values denote intensity DNs (40–120). Wet surfaces and surfaces without vegetation have low intensity. Gaps in the ground data are caused by canopy interception in the sparsely forested mire that surrounds the treeless mire (to the left and right). Reflections from the wavy water surface vary considerably in intensity.

The correlation ($r^2 = 0.59$ for the 28-cm pixels in direct light) implies that the geometry of the LiDAR and the images was well matched and that intensity data could constitute a complement to aerial images, especially in sparsely forested areas, where shadowing causes problems in image interpretation. Unlike spectral values in images, the intensity of rough surfaces is not greatly affected by the near-nadir viewing geometry of airborne LiDAR.

3.3. Supervised classification of mire habitats, using area-based LiDAR features

3.3.1. Feature extraction and investigation

The LiDAR features (Table 4) that we tested for the analysis of mire vegetation were inspired by those used in forest inventory applications for the assessment of timber resources and tree species (Holmgren and Persson, 2004; Næsset, 2004; Reitberger

Table 4

Features derived from the LiDAR data. We tried threshold values of 10–30 cm for the separation of ground and shrub layers. We applied 15 cm in all analyses.

Intensity features from first-return points	
<i>imt, isdt</i>	Mean and standard deviation (SD). Tree layer, $h > 1$ m
<i>ims, isds</i>	Mean and SD. Shrub layer, $h \in [0.15, 1$ m]
<i>img, isdg</i>	Mean and SD. Ground layer, $h < 0.15$ m
<i>int1, int2, int3, int4</i>	Mean intensity in quartiles of the relative height distribution
<i>intc1, intc2, intc3, intc4</i>	Mean intensity in height classes: <25%, <50%, <75% and $\leq 100\%$.
<i>i1–i10</i>	Intensity deciles 10, 20, ..., 100%, all returns
Features computed from point proportions	
<i>pt, ps, pg</i>	Proportion of first-return points. Tree, shrub and ground layers
<i>ps^{NORM}</i>	Normalized proportion in shrub layer points, $(1-pt) \times ps$
<i>p1, p2, p3, p4</i>	Proportion of first, second, third, and fourth returns
Height distribution features	
<i>hq1–hq10</i>	Deciles of the height distribution (m) 10, 20, ..., 100%, all returns
<i>hd1–hd10</i>	Deciles of the relative height distribution (%) 10, 20, ..., 100%, all returns
Features derived from the DEM and hummock-model, 20 m $\times$ 20 m squares used only	
<i>sdz</i>	SD of DEM values excluding a linear elevation gradient (see Section 3.1)
<i>rangez</i>	Range of DEM values excluding a linear elevation gradient
<i>humnum</i>	Number of hummocks
<i>humcomp</i>	Compactness of the hummocks
<i>humsize</i>	Mean size of hummocks
Transformations found useful in vegetation analysis	
<i>hd28, hq48, hq210</i>	$hd2/hd8 \times hq5, hq4/hq8, hq2/hq10$
<i>imgq5, intc24</i>	$img/iq5, int2/int4$
<i>iq25</i>	$iq2/iq5$

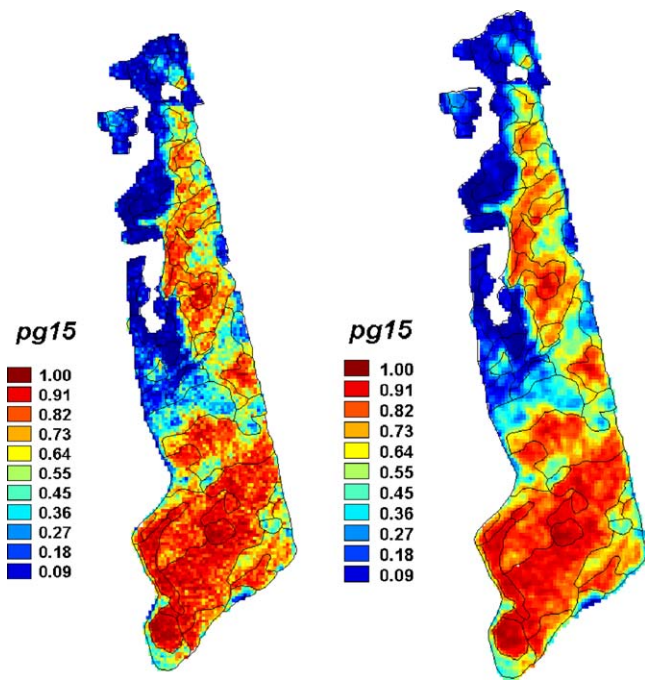


Fig. 8. Proportion of first-return ground points (*pg*, 15-cm threshold) in 10 m × 10 m and 20 m × 20 m squares.

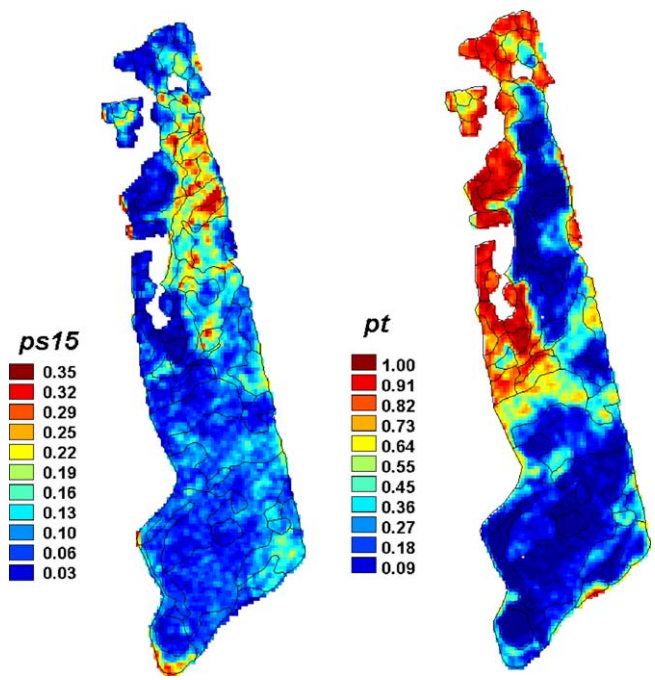


Fig. 9. Proportion of first-return shrub layer points (*ps*) of first-return tree canopy points (*pt*), 20 m × 20 m squares.

et al., 2006). We also tried to deduct new features that would be meaningful in mire habitat classification and that would measure the surface pattern of the mire as well as the density and type of understory and ground vegetation. For example, we hypothesized that the mean intensity of ground returns, feature *img*, could be used to characterize the ecohydrological conditions. Lakkasuo was divided into a 10 m × 10 m grid of 11,992 points and features were computed for 100-m² and 400-m² squares. The DEM was used for calculating the height (*h*) of each return. Table 4 lists the features used.

A total of 6705 data plots were located in pristine mire. We used these data and Box-Whisker plots, correlation analysis, ANOVA and visual interpretation of maps to examine the features for their explanatory power in mire habitat classification.

The proportion of tree canopy, shrub, and ground layer returns is shown in Figs. 8 and 9. As expected, variables *pg* and *pt* showed high negative correlation and a low proportion of echoes returned from the shrub layer, *ps*. Contrary to expectations, the classification of open mires with these features would be inaccurate, because of overlapping distributions (Table 5).

Treeless mires in Lakkasuo have scattered trees (*pt* > 0), especially the LkN-bogs. The ridge-hollow pine bogs (KeR), which have highly clustered spatial patterns of trees and large open areas (*pt* = 0), mix easily with treeless mires. Fig. 10 shows the pattern of *pt* in the ombrotrophic part of Lakkasuo. A high intraclass variation is evident. Mire site types change gradually and the mire site-type classification in the transition zones is ambiguous. If we assume that *pt* separates the ombrotrophic sites, the patterns in Fig. 10

would suggest that the reference site-type map was not correctly drawn.

The proportion of canopy and ground returns showed a strong relationship with site fertility in genuine forested site types (Table 6, Fig. 11).

The lowest values for *img* were observed at the herb-rich flark fen (RhRiN) sites. These fens are very wet with a high coverage of flarks. Forested spruce mires with high stocking had low values of *img*, due to canopy transmission losses in the first-return data or

Table 5
Percentage of LiDAR points in tree, shrub, and ground layers in pristine mire types according to the main type. Mean and SD. 10 m × 10 m squares.

Main type	<i>pt</i>	<i>ps</i>	<i>pg</i>
Treeless mires	4.0 (8.4)	15.3 (10.5)	80.7 (13.9)
Composite mires	26.7 (27.0)	10.4 (7.3)	62.9 (27.1)
Genuine forested	49.6 (34.0)	7.3 (5.9)	43.1 (32.0)
All, pristine	34.4 (33.1)	9.6 (7.7)	56.0 (31.2)

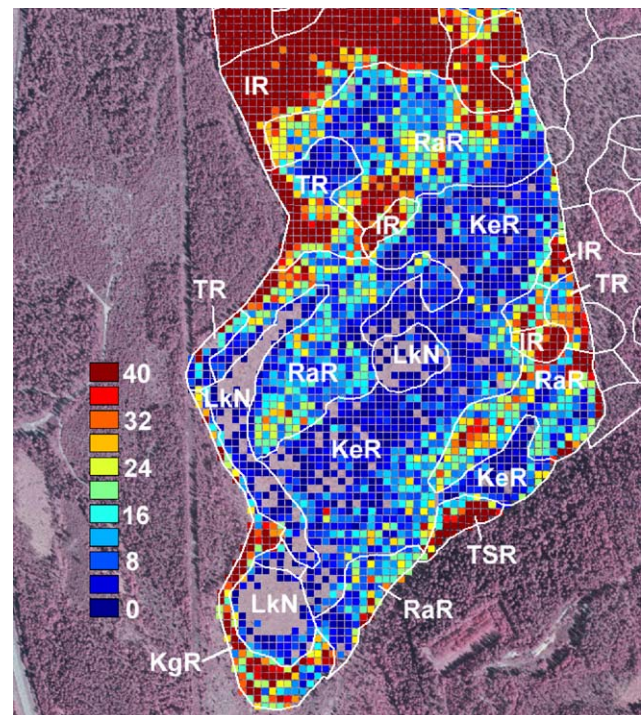


Fig. 10. Proportion of canopy returns (*pt*, here in %) in the ombrotrophic part of Lakkasuo. 10 m × 10 m grid.

Table 6

Percentage of LiDAR points in tree, shrub, and ground layers in genuine forested mires. Mean and SD according to the fertility classes. 10 m × 10 m squares.

Nutrient level	<i>pt</i>	<i>ps</i>	<i>pg</i>
ombro (RaR)	16.3 (13.1)	8.5 (4.5)	75.2 (13.7)
oligo-ombro (IR)	50.8 (16.7)	6.7 (3.8)	42.5 (15.3)
meso-oligo (PK, KgR, KR)	64.9 (24.8)	9.7 (7.5)	25.7 (20.8)
meso (MK, KgK)	86.3 (13.4)	4.9 (6.7)	8.7 (8.8)
meso-eu (RhK, LhK)	88.7 (11.3)	4.8 (6.5)	6.5 (6.3)

Table 7

Mean and SD of intensity of tree canopy points in genuine forested and composite mires according to dominant tree species. 10 m × 10 m squares.

Species mixture	<i>n</i>	<i>imt</i>	<i>isdt</i>
Spruce	566	31.7 (4.2)	15.3 (1.4)
Spruce–hardwood	761	31.4 (5.7)	15.5 (2.0)
Pine–hardwood	21	35.5 (2.9)	16.3 (1.1)
Pine–spruce	309	28.6 (3.1)	14.2 (1.2)
Pine	4025	27.0 (5.6)	11.2 (2.8)

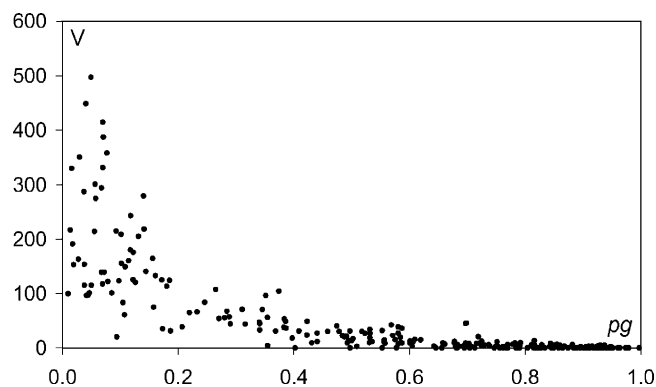


Fig. 11. Scatterplot of *pg* in 2006 and stem volume (m^3/ha) in Bitterlich plots in 1995 ($50 \text{ m} \times 50 \text{ m}$ grid). Pristine mire sites. $10 \text{ m} \times 10 \text{ m}$ squares.

actual differences in the ground vegetation. The differences in *img* between genuine pine-dominant and composite types were marginal. Contrary to expectations, the wet low-sedge bog sites (LkN) had the highest *img* (Fig. 12). It is likely that the sphagnum (*Sphagnum majus* (Russ.) C. Jens.–*Sphagnum balticum* (Russ.) C. Jens.) hollows were dry during the LiDAR campaign and thus showed high intensity. Then again, the hollows in the low-sedge and ridge-hollow pine bogs appeared bright in multitemporal color-infrared images.

In ANOVA, the mire site types explained only 8.9% of the variation in shrub layer intensity, *ims* (Fig. 12). The shrub layer features *ims* and *ps* did not capture the site-type-specific characteristics of this vegetation layer.

High values of *imt* occurred in squares comprised of alder and downy birch (Fig. 12). This result was verified by species determination in aerial images. The orientation and arrangement of alder leaves (resulting in large silhouette area) rather than the reflectance explains the higher intensity. The *imt* was higher in spruce canopies than in pine (Table 7). High intraclass variation implies that *imt* cannot be used for reliable determination of the tree species composition. The separation of pure spruce stands from stands with mixtures of hardwoods can be especially difficult. A more detailed analysis at the tree level is required.

We hypothesized that differences in the structure and species composition of the shrub layer of composite and treeless site types would be characterized by the features *ims*, *isds*, and *ps*. In ANOVA, 8.9%, 5.9%, and 36.5% of the variation in *ims*, *isds* and *ps* were explained by the mire site type, respectively. The proportion of shrub layer hits (*ps*) was highest in the minerotrophic treeless mires, which can partially be explained by the abundant tall sedge vegetation (Table 8). However, a likely reason for the high values of *ps* is the rough surface relief of these sites, details of which were not captured in the DEM.

The features derived from the height distribution were strong explanatory variables of the nutrient status, vertical structure of the forest canopy, and the tree species composition (Figs. 13–16).

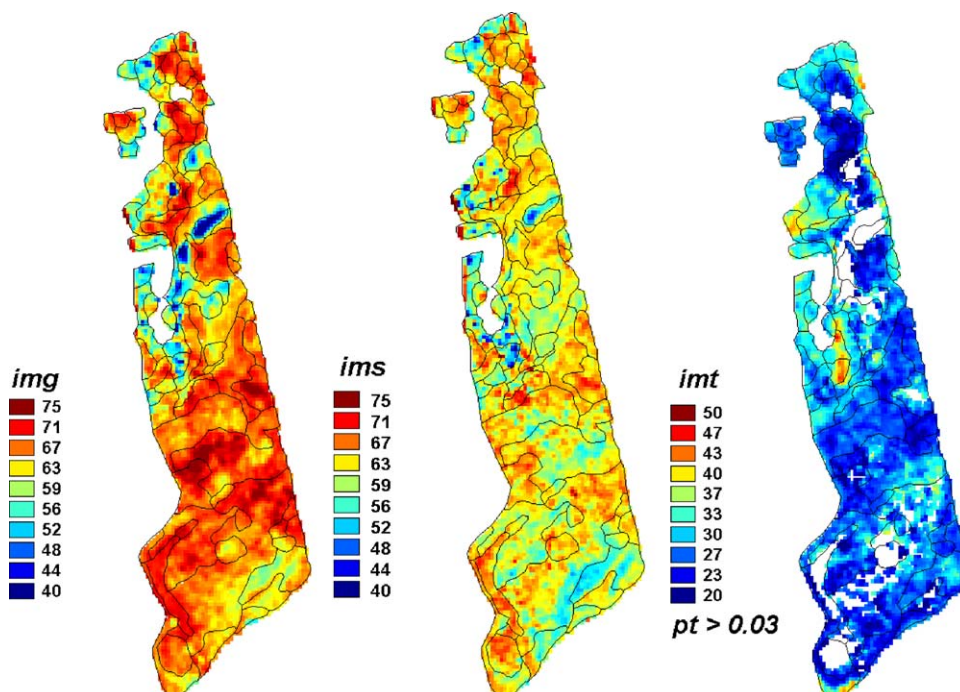


Fig. 12. Mean intensities of ground (*img*), shrub (*ims*), and tree canopy (*imt*) layers at pristine mire sites. *imt* is drawn for squares with $pt > 3\%$. $20 \text{ m} \times 20 \text{ m}$ squares.

Table 8

Mean and SD of proportion of shrub layer points (*ps*) in composite and treeless mires according to the nutrient level. 10 m × 10 m squares.

Nutrient status	Composite	Treeless
meso	9.2 (8.5)	20.6 (9.4)
oligo-meso	15.8 (8.0)	20.1 (9.2)
oligo-ombro	11.7 (7.4)	–
ombro	8.0 (5.0)	5.6 (4.4)

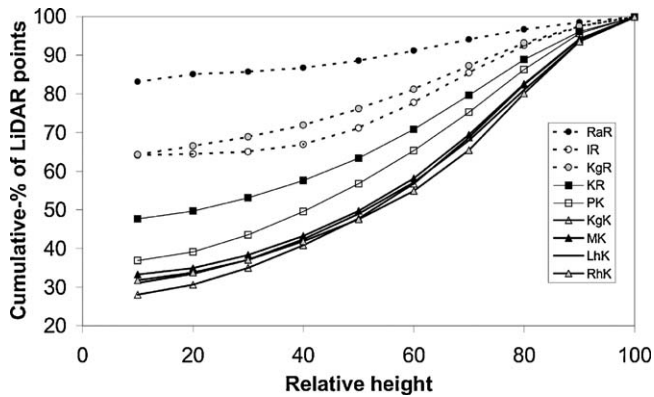


Fig. 13. Mean values of features *hd#* in genuine forested sites. The entries in the legend are in ascending order of nutrient level (~yield) level. Sites RaR, IR, and KgR are pine dominant. KR and PK are mixed pine and spruce. Spruce mainly occurs on KgK and MK sites. LhK and RhK are mostly spruce-dominant sites with mixed hardwood.

The dominant height (*hq10*) was highest in spruce mires, i.e. in the thin-peated sites: KgK and LhK (Figs. 15 and 16). The relative height distribution variables (*hd#*) were associated with site fertility, density, and vertical structure of the canopy and species mixture (Figs. 13 and 14). In dense canopies, a large portion of the LiDAR returns come from the upper canopy and a small proportion reaches the lower levels. In multilayered canopies (sites RhK, RhSK,

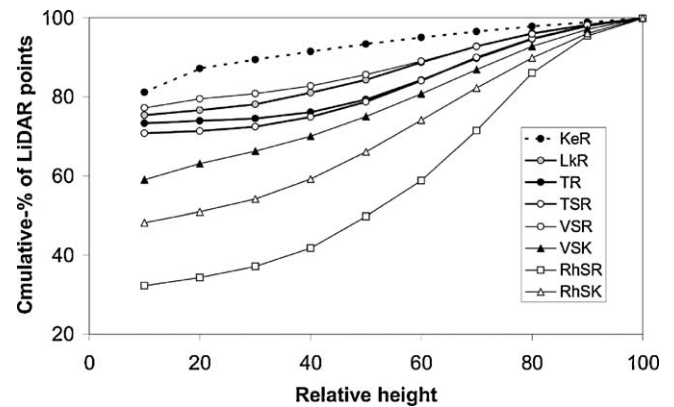


Fig. 14. Mean values of features *hd#* in composite site types. The site names in the legend follow the ascending order of nutrient level.

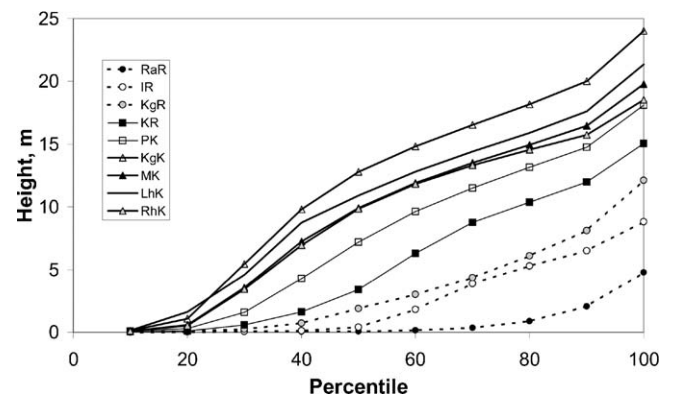


Fig. 15. Mean values of features *hq#* in genuine forested types.

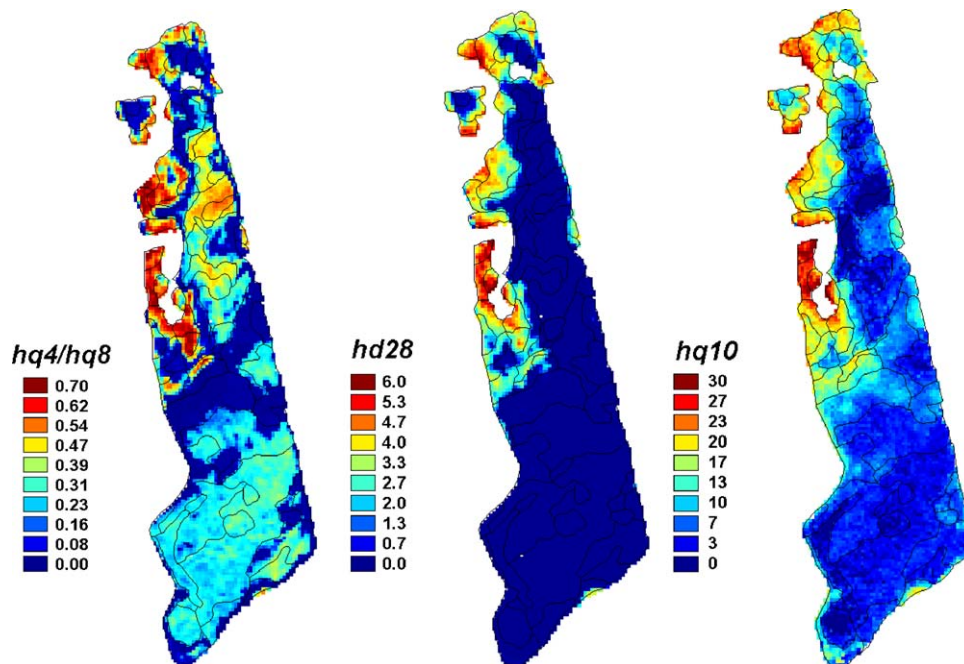


Fig. 16. Patterns of the transformed feature *hq4/hq8* grouped the sites according to the tree species and canopy density. The herb-rich fens and composite types also formed a group. Feature *hd2/hd8* × *hq5* (m) separated the spruce mires from other sites. Feature *hq10* measures the dominant height of trees (m). Maximum values of above 30 m were observed in the meso-eutrophic thin-peated spruce mires.

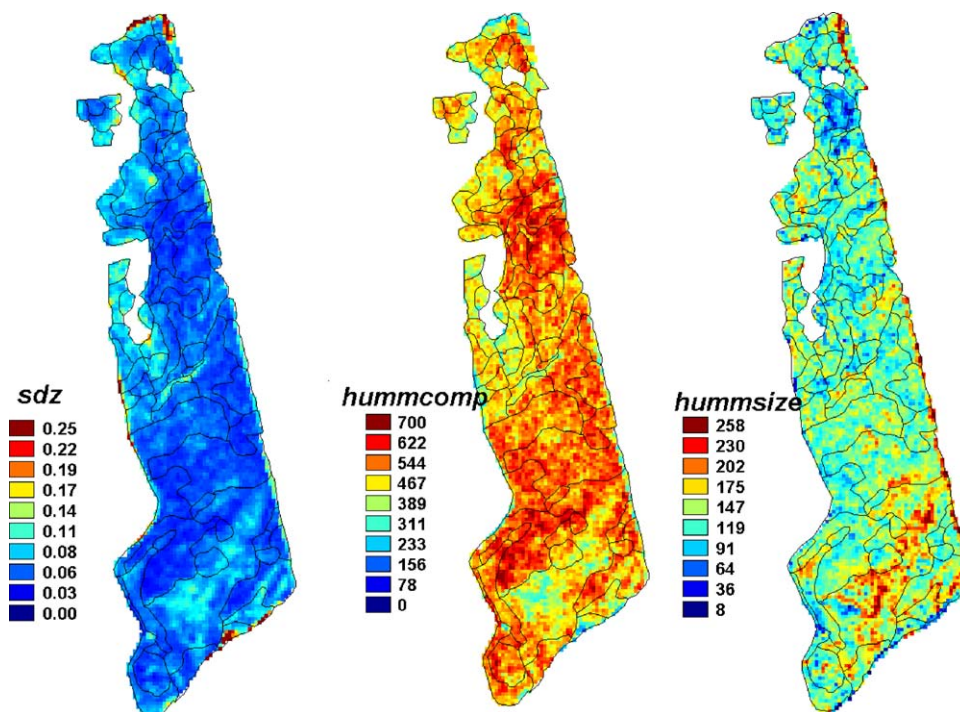


Fig. 17. DEM features *sdz* (m), *humcomp* (number of edge pixels), and *hummsize* (m²).

RhSR), echoes are registered from all height layers. In the pine-dominated bogs (RaR, KeR, IR, KgR, TR), canopy transmittance is high and the canopy returns have echoed from a single dominant tree layer (Figs. 14 and 15).

We tested transformations, using the *hd#* and *hq#* variables to determine whether the form of the cumulative height distributions would be useful in separating mire site types and found new variables that appeared visually promising (Fig. 16).

In genuine forested mires, the nutrient level explained 80% of the variation in *hq10*. In wet composite types, only 35% of the variation in *hq10* was explained by the nutrient level. Similarly, the site type explained 81% of the variation in *hq10* over all pristine site types, indicating high potential as an explanatory variable in mire classification. The DEM-based features behaved logically (Fig. 17) although they did not appear to be strong variables for separating mire site types according to visual inspection and ANOVA.

3.3.2. Classification methods

To examine the potential of LiDAR for mire habitat mapping and to determine the relevant features, we classified the data from the 10 m × 10 m and 20 m × 20 m squares in the pristine part of Lakkasuo according to Table 1:

- mire site type, 21 classes,
- main type, 3 classes (forested, composite, treeless),
- dominant tree species, 3 classes (spruce, pine, spruce–pine),
- nutrient level, 5 classes in forested mires and 4 classes in composite and treeless mires.

We used three classification methods including the Random Forest (RF), Support Vector Machines (SVM), and the k-Nearest Neighbor (kNN) methods with 63 explanatory LiDAR variables. RF could also be used for the selection of meaningful variables. We divided the data into training and testing sets, selecting every second square for each. This division represents an optimistic case with an ample training set. Thus, the classification results represent the upper bounds of accuracy achievable. The field plots ($n = 265$) were also used for training.

RF is a machine-learning ensemble classifier that consists of many decision trees and outputs the class that is the mode of the classes output by individual trees. It is based on the growing of numerous, typically hundreds or thousands of, classification trees, using randomly selected variables from the training set (Breiman, 2001). The randomForest 4.5–28 package in R (2.8.1) includes two measures of feature importance: the mean squared error effect and the Gini importance. The use of RF is technically simple because it requires only a few parameters. RF was used for forest structure quantification with LiDAR data by Falkowski et al. (2009) and for land-cover classification with multisource remote-sensing data by Gislason et al. (2005). A feature selection phase is not required with RF.

SVMs delineate two classes by fitting an optimal separating hyperplane to the training data in the d -dimensional feature space (Vapnik, 1998). They are based on structural risk minimization: a hyperplane is optimal when it minimizes a cost function that expresses a combination of two properties: maximization of the margin, i.e. the distance between the hyperplane and the closest training samples, and minimization of the error in training samples that cannot be separated (Bruzzone and Carlin, 2006). The influence of the nonseparable samples can be controlled by two different regularizations, either by C or ν -SVM parameterization. For linearly nonseparable cases, the input data are implicitly mapped into a higher dimensional space by a non-linear kernel function. SVMs were used for the classification of multispectral remote sensing combined with LiDAR by Alonso and Malpica (2008) and Koetz et al. (2008). We used the C -SVM in R with a linear kernel ($C = 1$).

The kNN method is widely used in remote sensing. It is a simple nonparametric method. The features form an n -dimensional space, where the classification decision of a test case is based on the classes of the nearest neighbors. We applied Euclidean distance and tested odd values of the parameter k . The best accuracy in mire site-type classification was achieved with $k = 9$, which we later applied in all kNN classifications.

We applied five different feature sets in all classifications that were derived using either the 10 m × 10 m or 20 m × 20 m data.

Table 9

RF and Expert variables in the feature sets used.

Classification	
RF-variables in Gini order of importance (20 m × 20 m)	
Main type	pt, hq10, hq1, hq48, imt, p1, hd6, hd5, p2, hq9, intc24, hq7, sdz, hq6, hd4
Mire site type	hq10, hq9, hq1, sdz, hq5, hd28, hq2, hq4, hq8, hq3, pg, hq6, img, hq48, pt
Dominant tree sp.	hd28, hq5, hq4, p3, hd1, hq3, pg, hd2, hq6, hq9, hq8, hq2, hq7, img, int1
Forested mires	hq10, hq9, p3, hq5, pg, hq6, hq8, hd28, hq7, hd1, hd2, iq1, hq4, hd3, hq48
Composite types	hq1, ps, hd28, hq3, hq2, ps _{NORM} , hq5, hq4, hq6, pg, hq48, hq10, hq7, sdz, int1
Open mires	hq1, hq2, hq4, hq3, hq5, hq6, ps, hq7, hq8, ps _{NORM} , hq9, pg, intc4, hq48, intc2
Expert variables (20 m × 20 m)	
Main type	hummsize, sdz, pg, ps, pt, p1, p2, hd1, hd5, hd28, hq48, intc24, hq210, iq25, imgq5
Mire site type	hummsize, sdz, pg, ps, pt, p1, p2, hd1, hd5, hd28, hq48, intc24, hq210, iq25, imgq5
Dominant tree sp.	hd1, hd3, hd5, imt, int4, hq210, hd28
Forested mires	pt, imt, intc24, hq210, iq25, hq48, hd28
Composite types	pt, ps, pg, imt, intc24, hq20, iq25, hq48, hd28
Open mires	ps, pg, p1, p2, hummnum, imgq5, iq25, intc24

The sets were referred to as Full, RF variables, PCA circle, Expert and PC. All variables were included in the Full set and were used with the RF and SVM classifiers. The 15 most important variables, as measured by the Gini attribute, were selected in the set of RF variables. The PCA circle variables, 15 in total, were selected manually using a variable correlation circle plot by the PCA procedure in R (Package FactoMineR v. 1.10). These were *iq1*, *iq5*, *iq8*, *int1*, *intc4*, *p1*, *p2*, *p4*, *pt*, *hq3*, *hq5*, *hd6*, *intc24*, and *hq48* for both square sizes. We also selected 15 Expert variables based on our mire ecological knowledge, analysis of Box-and-Whiskers-plots, and maps of individual features, as well as correlation clustering of the variables. A different set of Expert variables was chosen for each classification (Table 9). The set PC consisted of the first 15 principal components.

We also tested feature subsets that intentionally excluded variables that could lead to overfitting or problems with generalization of the results. For example, the metric *hq*-height features can be problematic, because the forested mire site types at

Lakkasuo represent only minor variations in forest/mire succession. Overall classification accuracy and the simple kappa coefficient were used as performance measures. Kappa is +1 when there is perfect agreement and zero when a random assignment of the classes would have been as effective as the applied classifier (Cohen, 1960).

3.3.3. Classification results

The classification results are in Table 10.

3.3.3.1. Mire site types. A total of 21 mire site types were represented in the reference map. Due to the subjective nature of the classification and the fact that some of the site types strongly resemble each other in the field, we also measured the performance by allowing for minor errors (Table 11). For the 20 m × 20 m data, the accuracy of mire site-type classification was 33–66% ($\kappa = 0.25$ – 0.62), depending on the method and variables. Omitting minor errors, the accuracy was 64–78%, without changes

Table 10

Classification accuracy (%) and simple Kappa for the 20 m × 20 m (left) and 10 m × 10 m cases (right).

Method and feature set	Main type				Mire site type				Dominant tree sp.			
	%	κ	%	κ	%	κ	%	κ	%	κ	%	κ
SVM Full	70	0.50	67	0.45	60	0.56	51	0.46	90	0.76	88	0.72
RF Full	80	0.67	74	0.57	66	0.62	52	0.46	92	0.81	90	0.75
kNN rf-vars	65	0.42	60	0.32	45	0.39	33	0.24	88	0.72	88	0.70
SVM rf-vars	61	0.36	61	0.35	52	0.47	45	0.38	90	0.71	87	0.69
kNN PCA-circle	66	0.48	62	0.37	45	0.39	37	0.30	88	0.69	87	0.67
SVM PCA-circle	61	0.35	61	0.34	47	0.41	42	0.35	88	0.70	88	0.70
kNN Expert	59	0.32	59	0.31	33	0.25	31	0.23	89	0.72	87	0.69
SVM Expert	65	0.42	62	0.37	50	0.44	45	0.38	88	0.71	87	0.67
RF Expert	75	0.59	71	0.51	59	0.55	48	0.42	90	0.75	89	0.72
kNN PC	70	0.51	66	0.43	49	0.43	42	0.36	88	0.72	86	0.66
SVM PC	63	0.37	61	0.35	50	0.45	34	0.27	87	0.69	87	0.67
Nutrient level												
	Forested mires				Composite types				Open mires			
	%	κ	%	κ	%	κ	%	κ	%	κ	%	κ
SVM Full	79	0.72	75	0.66	86	0.74	78	0.59	77	0.65	69	0.53
RF Full	83	0.77	75	0.66	86	0.75	79	0.61	81	0.71	75	0.62
kNN rf-vars	72	0.63	69	0.59	72	0.47	68	0.41	65	0.47	65	0.47
SVM rf-vars	74	0.64	71	0.61	81	0.66	76	0.55	79	0.66	72	0.57
kNN PCA-circle	72	0.62	67	0.55	70	0.43	65	0.33	67	0.50	62	0.42
SVM PCA-circle	75	0.66	72	0.63	76	0.54	69	0.39	78	0.67	67	0.50
kNN Expert	68	0.56	64	0.50	69	0.41	66	0.33	66	0.48	59	0.38
SVM Expert	70	0.59	67	0.55	77	0.56	74	0.51	68	0.50	71	0.56
RF Expert	77	0.69	71	0.61	81	0.66	75	0.53	76	0.63	69	0.53
kNN PC	71	0.60	68	0.57	76	0.53	75	0.54	77	0.65	64	0.45
SVM PC	74	0.65	64	0.51	79	0.61	66	0.39	78	0.66	69	0.53

Table 11

Confusion matrix of SVM mire site-type classification using Expert variables from 20 m × 20 m squares. Classification accuracy was 50%. The gray cells denote minor errors in site-type classification. Allowing for minor errors, the accuracy was 70%.

	Genuine forested types									Composite types								Treeless mires				Classification accuracy, %	
	Spruce mires					Pine mires				Minerotrophic				Ombrotrophic				Minerotrophic		Ombrotrophic		With minor errors	Correct cases only
	RhK	LhK	MK	KgK	PK	KR	KgR	IR	RaR	RhSK	RhSR	VSK	VSR	LkR	TSR	TR	KeR	RhSN	RhRiN	VSN	LkN		
RhK	69		45	5		1	9			5			2									84	51
LhK	5	6	16	6			2			1												75	17
MK	24		126	17	4	1	11	4		1		4	2		2							87	64
KgK	8	1	31	41	2					1					1							87	48
PK	11		45	2	10		1	2		2		2	2									75	13
KR	1		7		2	16	7	4	19	1		7	23		3	4	2	2				26	16
KgR			20		2		30	17		2			8									62	38
IR			2					146	51				3		2	22						74	65
RaR								21	452				17		4	22	111	4	1	2	1	75	71
RhSK	17		14	1	4		6	1	8	25		13	16				1					36	24
RhSR	1		3							5	0		1									10	0
VSK	2		11		5		7	1	7	8		16	38			2	1			1		24	16
VSR	1				3		5	5	65	4		6	148		4	6	6	3		1	7	58	58
LkR								2	5					0	1		3					45	0
TSR			5		1		1	1	13	1		5	17		12	1						53	23
TR			8			1		56	111			5	16		2	66	9			2		85	24
KeR								1	125						6		375	1			1	74	74
RhSN									13			4	22				14	10		26		40	11
RhRiN									7			6	8				1	7	31	25		74	36
VSN			3		1			2	13			1	16		1		3	13	9	67		69	52
LkN									11							1	112			1	24	92	16

in the ranking of methods. RF produced the most accurate results and kNN with the Expert variables led to the lowest accuracy (Table 10, Fig. 18). The accuracy was lower with the 10 m × 10 m data, 31–52%, but when omitting minor errors, 53–72%.

The RF variables were mostly *hq#* and *hd#* features. The absolute height of the trees and the vertical canopy structure separated the mire site types. These features are used in field classification, at least implicitly. Since the species of ground vegetation, which are carefully examined in the field, cannot be

characterized by the LiDAR features, the height-distribution variables became the best available explanatory variables. The bottom flora may have affected the mean intensity of the returns from below 15 cm, *img*. In the 100-m² data, the DEM variables *humcomp* and *rangez* were also selected by RF, which may indicate the increase of noise in the point-based features in the smaller plots. The 400-m² data outperformed the 100-m² data. The features in the smaller plots may have been prone to noise, due to the smaller number of pulses. In some site types, the site-type specific characteristics were likely not captured by the 100-m² plots. It should be noted that the neighboring 20 m × 20 m squares overlapped the 10-m grid of the center points. The 20 m × 20 m squares of the training and test sets thus overlapped, which evidently resulted in optimistic results.

Use of the absolute height distribution percentiles can produce optimistic results that cannot be generalized outside Lakkasuo, where the forested mires have long escaped cuttings and forest fires and are rather evenly aged. In disturbed mires, e.g. mires that are in the process of restoration, it may be advisable to use those variables that characterize the canopy density and the shape of the height distribution. RF was used without the *hq* features and the accuracy decreased to 60% ($\kappa = 0.55$). Since the *hd* variables separated the forested and composite site types (Figs. 13 and 14), we tried RF using only the *hd* variables; the resulting accuracy was 43%. We also tried classification, using the field plots ($n = 265$) for training. The RF variables remained, but the accuracy of RF decreased to 46% and that of kNN with the RF variables to 34%. These values give a more realistic indication of the accuracy achievable in mire site-type classification using LiDAR. The results of Table 9 can be achieved, using extensive training data. Changing the amount and layout of the *in situ* training data affects the results of supervised classification.

The site type with the highest accuracy, 60–79%, was the *Sphagnum fuscum* pine bog (RaR), which is the most common site type in the Lakkasuo mire. Among the common site types, most errors were made in the tall-sedge types. For example, the tall-sedge pine fen (VSR), which differs highly in ground flora, was

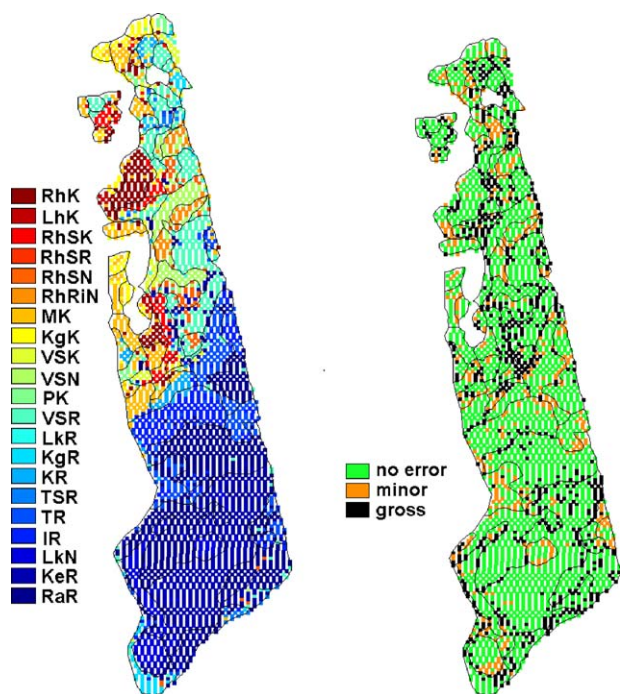


Fig. 18. Predicted mire site types in the test set and a 3-class error map. RF with the Full feature set. 20 m × 20 m squares.

Table 12

Confusion matrix of the kNN classification, using Expert features and the 20 m × 20 m data.

Main type	Predicted			
Observed	Treeless	Composite	Forested	All
Treeless	203	169	80	452
Composite	109	703	520	1332
Forested	39	447	1082	1568
All	351	1319	1682	59.3%

mixed with the *Sphagnum fuscum* pine bog (Table 11); both have 3–8-m high pines. In all, errors were found near the boundaries of the reference compartments (Fig. 18), which is consistent with site types being simplifications of the natural gradients found in mires.

3.3.3.2. Main types. The classification accuracy of the main types was 59–80% ($\kappa = 0.32$ – 0.67). The most important RF variable was the proportion of canopy layer points, *pt*. The key difference between the main types is by definition the amount and spacing of trees. The *hq* variables and the proportions of first- and second-return points were important explanatory variables. The forested mires were often confused with the composite types (Table 12). The treeless mires were also mixed with the composite types. The error rates of 20–40% exceeded our expectations, since we had anticipated very high accuracy for the detection of open fens and bogs. The open mires of Lakkasuo have scattered trees and some of the most barren forested and composite sites have large openings (KeR, RaR).

3.3.3.3. Dominant tree species. The RF variables excluded intensity features apart from *intc24*, which is the ratio of the mean intensities of all points and points in the lower half of the height distribution. The Expert variables included *imt* and *int4*, because these showed explanatory power (Table 7, Fig. 12). The crown shape and depth are different in pine and spruce, which affects the shape of the relative height distribution (Fig. 13). The spruce sites had higher canopy absorption and growing stock and the trees are taller. Spruce sites are also characterized by a more abundant understory vegetation that consists of downy birch and spruce; this may affect the values of *hd1*–*hd3*. The classification accuracy was 87–92% and all methods produced comparable results. When we tried RF without the *hq* variables, the accuracy was 89%. In this case, the most important RF variables were the *hd* variables. With six canopy intensity features, the accuracy was 88%, 85%, and 87% using RF, SVM, and kNN, respectively ($\kappa = 0.66$ – 0.72). The descending order of Gini importance was *isd1*, *iq25*, *imt*, *int3*, *int4*, and *int2*. These results suggest that intensity features may be used for the separation of pine and spruce, which are difficult to differentiate in aerial images.

3.3.3.4. Nutrient level. Classification was performed separately for forested, composite, and open mires. Again, the *hq* variables were important by RF. They separated the forested types, which is consistent with the trees on the dry forested sites at Lakkasuo having mostly reached the maximum attainable height. The proportions of shrub layer points were important explanatory variables in the composite and open mires. The nutrient-rich open site types at Lakkasuo have a vigorous shrub layer (*Betula nana*, *Salix* sp.) and the microtopography is rough. Each may have increased the number of shrub layer returns in open sites. Indicators of nutrient status in composite and open mires are, however, formed by ecological plant groups prevailing on lower and wetter surfaces. The tests carried out at the Silmäpäänlammi fen confirmed that these species groups (excluding eutrophic vegetation) cannot be separated based on LiDAR intensity and the

Lakkasuo results confirmed this. The proportion of shrub-layer points was small at the forested sites. The shrubs, e.g. *Rhododendron tomentosum*, *Vaccinium uliginosum* and *Betula nana* in the pine bogs, did not give rise to a reflection 0.15–1 m above the DEM. The proportion of third-return points was important, since it measures the depth and transparency of the canopy.

4. Discussion

4.1. Confines of this study

We did not test for varying LiDAR footprint or pulse density. An increase in point density would likely improve the accuracy of mire surface elevation modeling. The features derived from the intensity and height distributions would also likely be less noisy. Changing the footprint size would likely influence the height distributions, but differently for tree canopies and other short canopies, as shown by Hopkinson (2007). The intensity signatures of tree species would probably change if the footprint is altered. The actual target reflectance contributes more in intensity for small footprint sizes that approach the size of the scattering elements, i.e. shoots and leaves. If a higher pulse density is achieved by means of a higher pulse repetition frequency, it may skew the point height distribution due to a reduced signal-to-noise ratio (Hopkinson, 2007). *In situ* calibration and training data are therefore needed if the LiDAR parameters are changed between mire sites and LiDAR campaigns.

We did not test the accuracy of LiDAR in retrieving tree height (accuracy of *hq/hd* variables). Trees in ombrotrophic bogs and wet mires have sparse foliage, which may affect this accuracy and canopy heights are probably underestimated more. Differences in the needle density and mass due to site conditions may also show as differences in the canopy intensity metrics.

One discrete-return LiDAR sensor was used. The use of full-waveform LiDAR would probably have led to enhanced results, since it provides the width and amplitude of the echo. The ALTM3100 and other discrete-return systems only use the amplitude, because intensity is believed to correspond to it (exact details by the manufacturers are lacking). Echo width can be exploited in tree species determination (Reitberger et al., 2006) and helps in the classification of ground hits (Ullrich et al., 2007). Waveform LiDAR could potentially be used for better separation of low vegetation and mosses.

The results concerning the response of mire vegetation to LiDAR intensity may have been influenced by the dry conditions that preceded the LiDAR campaign. The normally wet hollows in the ombrotrophic rain-fed sites were especially dry and the pale peat moss surfaces gave a signal different from that of a wetter period.

Our results in supervised habitat classifications were obtained, using abundant training data. We interpreted them as the upper bounds of accuracy achievable. In a more realistic scenario, in which 265 field plots were used for the training, a noticeably lower accuracy of mire site-type classification was obtained. This exemplifies the effect of the *in situ* calibration data in supervised classification. We performed experiments at two sites. It is evident that they represent incompletely the variation in important mire features. Our reference mire site-type map contained errors resulting from the interpolation method by which it was derived and the subjective, inexact nature of the mire site-type classes. We estimated that a maximal accuracy of 85–90% would have been possible, considering the map errors and the ambiguities of the transition zones.

4.2. Mire surface modeling using LiDAR

Our results suggest that LiDAR enables accurate 0.05–0.1-m mapping of mire surface elevation, at least when peat moss

bottoms prevail and the tree canopy is sparse. The sedge and shrub vegetation lifted the echoes 2–5 cm above the moss surface. The density of the LiDAR data and the footprint influences the accuracy. Calibration of systematic errors in the LiDAR trajectory and range measurements is necessary, as well as careful matching of overlapping strips, if very unbiased DEM data are essential (Csanyi et al., 2005). This calls for accurate *in situ* measurements. To maximize the quality and number of ground points for the DEM, it may well be advisable to use a semiautomatic approach with user intervention and local adjustment of the parameters of ground-point filtering. We did not try different algorithms or their parameters, but used a standard product by the vendor. We derived a binary hummock map, using the LiDAR DEM. Our simple approach omitted the intermediate levels between hummocks and hollows, which is clearly a shortcoming. If dense LiDAR data were available and a more advanced algorithm were used, we believe that hummocks, hollows, flarks, and intermediate levels can be localized at an accuracy that is relevant for monitoring applications. We tested the hummock model by superimposing it on aerial images and by using its features for habitat classification. The features were not strong explanatory variables, which may mean that the model was inaccurate or that the features were deduced poorly. The patterns in the images matched flawlessly, however. It should be recalled that a DEM is somewhat ambiguous in the flark, mud-bottom, and open-water areas that prevail, e.g. in aapamires. During work with the hummock model and the DEM, we noted that the ditches in the drained part of Lakkasuo were clearly visible in the DEM. Using LiDAR for monitoring the status of the drainage network seems feasible. High sampling rates are likely required, as well as leaf-off data.

In considering the monitoring of mire surface profiles with LiDAR, one should consider the factors affecting the topological profile of a bog at different time scales. The so-called bog breathing (Mooratmung), caused by accumulation and release of methane gases, occurs in hourly cycles and the effect on surface height can be over 20 cm (Glaser et al., 2004). Rise and fall of the mire surface according to water level has a timescale from days to months, depending on rainfall, and the effect can be several centimeters between periods of rain (Glaser et al., 2004). The accumulation of peat in a typical boreal bog occurs at rates of 0.2–4 mm/yr (Korhola and Tolonen, 1998).

4.3. LiDAR features in the characterization of mire vegetation and habitats

We tested the intensity response of 0.25-m² samples of open-mire vegetation and a total of 63 features that were derived from pulses in the 100-m² and 400-m² squares from a diverse mire. Analysis with accurately mapped vegetation samples showed that surface moisture affects intensity. We observed only minor interclass differences in intensity for the plant communities tested, representing different nutrient levels and consisting of peat mosses, sedges, and herbs. The intraclass variation was high, even after we normalized the intensity data for the varying sensor-to-target range. Wet flarks and mud-bottom surfaces showed low intensity and our results suggest that these could be separated reliably. The 1064-nm intensity of the LiDAR correlated strongly with the NIR aerial image data. This suggests that LiDAR intensity complements optical data, especially in sparsely forested composite mires, where shadows hinder image interpretation.

We hypothesized that the first-return points in the shrub and ground layers, characterized by low transmission losses, would differentiate the site-specific lower flora. This hypothesis was not supported by the results. The features describing the canopy height profile and density, as well as the dominant tree species, were the

key variables in the mire ecological classifications. The composition of ground vegetation was out of reach with LiDAR, even in treeless mires. The geometric features that characterized surface patterns were surpassed by other features. The tree species mixture was classified by features that describe the shape of the relative height distribution and intensity of canopy hits. These results are in line with results in single-tree remote sensing in mineral soils (Holmgren and Persson, 2004).

The amount of the training data greatly affected the classification accuracy. It was interesting to note that the boundaries found, using the 265 plots as training data, were in agreement with gradients visible in the aerial images. This stresses the point that mire site-type classification is an approximation of a smaller-scale diversity and always subjective, depending on the classifier's ability to distinguish small and sometimes covered but (for ecohydrological classification) important moss species (Cajander, 1913; Laine and Vasander, 1996, 2008; Rydin and Jeglum, 2006).

4.4. Classification tools and feature selection methods

We used several feature sets, because the selection of an optimal set of explanatory variables from 63 is ambiguous. Similarly, we applied three classifiers to control their differences. RF produced highest classification accuracy and was superior in the classification of the 21 mire site types. In dominant tree species classification, there were only minor differences between RF, SVM, and kNN. RF and SVM modeled the complex relationships better than did kNN. In addition to the linear kernel, we tried SVM with polynomial and radial kernels, but only minor differences were observed. We tested four feature selection procedures with the kNN and SVM to examine the sensitivity of the results. Quite similar results were obtained independently of the feature selection procedure: automatic selection by RF, use of PCs, and the two subjective methods. The use of automatic selection by RF is an attractive option, because an exhaustive review of the individual features is laborious. On the other hand, careful visual examination of the features proved to be valuable for error detection. Because of the randomness of RF, the set of 15 RF variables varied between runs. In a trial of 10 runs, 18 different variables were listed. The most important features were stable.

4.5. Applications of LiDAR in mire environments

Based on our results, airborne discrete-return, small-footprint LiDAR offers a tool for accurate mapping and long-term monitoring of the mire surface and tree canopy. We tested LiDAR in classification tasks with mire ecological relevance. Spatiotemporal changes in mire vegetation are, however, continuous. We see future possibilities for the monitoring of growth, recruitment, and mortality of trees in treeless and composite mires. Monitoring of restoration sites should also be possible, using change detection methods (Yu et al., 2008). The actual implementation of a mapping and/or monitoring system would likely combine fieldwork for *in situ* calibration and training data, airborne images, and LiDAR. The use of LiDAR for prestratification and optimal allocation of field plots would rationalize the work. Based on our results, monitoring of restoration sites, in which restoration aims at diversifying the canopy structure to resemble that of the original mires, could be carried out using LiDAR as an aid. The recognition of dead standing trees in images and LiDAR would also be beneficial. The monitoring system should be robust to allow temporal changes in the LiDAR and field data acquisition processes. For example, we used intensity features and features describing the proportions of returns. The intensity data and point height profiles can change

notably if the LiDAR data acquisition settings are altered (Hopkinson, 2007). The return proportions are even more dependent on the sensor. Invariant features should be favored or the data and features should be calibrated in each campaign. Empirical allometric biomass or classification models produce biased results if unstable explanatory variables are used, or the LiDAR settings between sites do not match. *In situ* calibration is thus required, in which natural or artificial targets are used for normalizing the multitemporal datasets or for absolute calibration. Calibrating the geometry, e.g. for the inherent underestimation of tree heights, is also required.

4.6. Suggestions for future research

Our results suggest that LiDAR intensity offers potential for species recognition and monitoring of changes in the species mixture of trees in mires. More reliable results are obtained if the species identification process is carried out at the crown level to minimize the effect of interclass noise in the intensity data, which is present in plot- or area-based features. Reliable species recognition would improve the accuracy of habitat classification, although it can be challenging in multilayered stands, because fewer observations are always available for the suppressed trees. We did not include tests of the ability of LiDAR in measurement of the dimensions and biomass of the trees, since there are results available that treat these issues (e.g. Næsset, 2004; Packalén and Maltamo, 2007). The accuracy at which the poorly grown trees in the most barren or wet sites can be measured with LiDAR has not been examined and constitutes a future research topic. We also suggest research in tree species identification in mire forests, where the effect of site-type fertility should be examined. The accuracy of LiDAR in characterizing the vegetation of open mires, especially, could perhaps be improved by combining aerial images and LiDAR (Koetz et al., 2008; Korpela et al., 2008). We also propose research in the use of full-waveform LiDAR for the classification and characterization of mire habitats. Echo width and amplitude could potentially be used for better characterization of canopy and mire surface returns, resulting in improved DEM accuracy, classification of tree species, and characterization of the ground flora.

5. Conclusions

LiDAR succeeded well in characterizing tree species. *Alnus glutinosa* especially was well distinguished. Site fertility may affect specific tree species intensity characteristics. Surface characteristics of mire site types could be delineated with high accuracy. This offers excellent modeling possibilities for hummock-hollow stratigraphy and its development.

LiDAR failed in characterizing the differences in ground flora where species are small but crucial in the determination of the nutrient status of mire site types. Mire site types were modeled by the height distribution features. Differences in the wetness of the mire surface have a strong influence on intensities that should be considered in interpreting the results. LiDAR can be applied for the mapping and survey of tree stands on mires and could offer data that did not exist earlier for biotope mapping and investigations.

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